

# ELI — Architecture

ELI v2.3.1

August 2026

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## Blueprint — ELI MKXI Architecture

The full structural/architectural map of ELI MKXI. Every claim here is grounded in the actual source tree (file paths are real and clickable). Where something is observed-at-runtime rather than read-from-code it is marked (*runtime*).

ELI is a **local-first, offline-by-default, model-agnostic** cognitive runtime + assistant GUI (and, as of 2026-06-28, a self-hosted web app — `api/server.py`, documented in `ELI_USER_MANUAL.md`). No cloud, no APIs on the inference path, no hardcoded model. ~160k LOC across 397 Python files (+ `api/server.py`); 216 capabilities (2026-08-06).

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### 0. Design principles (the non-negotiables)

1. **Local-first / offline-by-default.** All outbound network is fail-closed at the socket boundary (`eli/core/netguard.py`); networked features opt in.

2. **Model-agnostic.** No model name/size is hardcoded on the inference path; ELI discovers and loads whatever GGUF exists. The model is swappable substrate, not a constant.
3. **Grounded / anti-confabulation.** A large dedicated subsystem exists purely to stop a local model stating things it can't back up (see §9).
4. **Emergent persona.** ELI has a distinctive voice; persona drift/banter is intended. Functional bugs are fixed; the personality is not scripted away.

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## 1. Repository map (recursive LOC, role)

*Per-package files / LOC measured 2026-07-28.*

| Package        | LOC   | Files | Role                                                                                                      |
|----------------|-------|-------|-----------------------------------------------------------------------------------------------------------|
| eli/runtime    | 30.4k | 94    | Grounding spine, evidence, re-sponse/introspection surfaces, daemons, self-improvement                    |
| eli/execution  | 24.2k | 15    | Router + executor: intent → action → side effects (executor god-file)                                     |
| eli/gui        | 22.7k | 24    | PySide6 desktop app, panels, startup/first-boot wizard, animated face                                     |
| eli/cognition  | 15.4k | 31    | Agent bus, 12-stage orchestrator, inference, persona, reasoning, tone/emotion adaptor, emotional timeline |
| eli/kernel     | 14.7k | 8     | CognitiveEngine (the orchestrating core), scheduler, task bus                                             |
| eli/perception | 8.3k  | 23    | Vision (VL), STT (whisper), TTS, <b>wake word</b> , <b>voice/tone</b> , OS control, gaze                  |
| eli/core       | 8.1k  | 28    | netguard, paths, settings, hardware profile, model download, <b>full_control</b>                          |
| eli/tools      | 7.4k  | 29    | Image engine, news, registry/capabilities, document tools                                                 |

| Package               | LOC          | Files      | Role                                                                    |
|-----------------------|--------------|------------|-------------------------------------------------------------------------|
| eli/memory            | 7.4k         | 13         | SQLite + FTS5 + FAISS + knowledge graph + working memory                |
| eli/planning          | 4.0k         | 24         | Proactive daemon, habit scheduler, job/proposal/attention queues        |
| eli/learning          | 3.4k         | 12         | LoRA self-training pipeline (Phi-3 base), dataset build/eval            |
| eli/coding            | 2.1k         | 12         | CodeAgent — plan→search→verify→repair coding pipeline                   |
| eli/plugins           | 2.0k         | 26         | Plugin manager + bundled plugins                                        |
| eli/world             | 1.6k         | 26         | EliWorld event bus, local world bridge, avatar/ontology/face            |
| eli/integrations      | 1.1k         | 9          | Optional Ollama client + integration adapters (not on the default path) |
| eli/utils             | 1.0k         | 3          | logging, shared helpers                                                 |
| eli/contracts         | 0.7k         | 3          | typed pipeline contracts                                                |
| eli/system            | 0.3k         | 2          | system-level helpers                                                    |
| eli/cli               | 0.1k         | 2          | headless REPL                                                           |
| <b>total (listed)</b> | <b>~155k</b> | <b>383</b> | full eli/ tree: ~160k / 397 incl. small top-level pkgs                  |

### The four god-files (refactor targets — see §20)

| File                                        | LOC   |
|---------------------------------------------|-------|
| eli/execution/executor_enhanced.py          | 14951 |
| eli/kernel/engine.py                        | 13841 |
| eli/gui/eli_pro_audio_gui_v2_0.py           | 12100 |
| eli/execution/router_enhanced.py            | 7621  |
| eli/runtime/deterministic_grounding_gate.py | 4301  |
| eli/memory/memory.py                        | 4734  |

## 2. Entry points & launch

| Command                       | Target                        | Notes                                       |
|-------------------------------|-------------------------------|---------------------------------------------|
| eli, eli-mkxi, eli-cli        | eli.gui.app:main              | GUI (default)                               |
| eli-gui                       | eli.gui.app:main              | GUI script                                  |
| python -m eli                 | eli/__main__.py               | dispatches to GUI or headless               |
| python -m eli --headless / -H | eli.cli.headless:run_headless | terminal REPL, no Qt                        |
| ./eli.sh                      | python -m eli                 | venv launcher                               |
| bash install.sh               | —                             | venv + deps + clean config seed<br>+ verify |

Boot path (GUI): app.py → (first-boot wizard if no model, eli/gui/panels/startup.py) → eli\_pro\_audio\_gui\_v2\_0.py:main  
 builds EliMainWindow → constructs the **CognitiveEngine singleton** → loads GGUF per config/settings.json  
 → starts daemons.

### 3. The request lifecycle (the spine)

Everything funnels through **CognitiveEngine.process()** (eli/kernel/engine.py). Two paths diverge by reasoning mode:

```

user input
  |
  ▼
[Router] eli/execution/router_enhanced.py :: route()
  |   priority pipeline → {action, args, confidence, meta.matched_by}
  |   ▼
CognitiveEngine.process(action, args, ...)
  |
  ├── PHASE45 fast-path? (deterministic OS/media/status/job actions)
  |   → execute_action() → return VERBATIM (no LLM)           [$10]
  |
  ├── non-quick mode OR forced → Internal Orchestrator (full 12-stage) [$7]
  |
  └── quick mode (default) →
      AgentBus.dispatch() [$6] → bus_result {grounding, agents, ctx}
          |
          ├── Grounding escalation hook [$9] (CHAT + low grounding + factual)
          |   → web tier / local tier / hedge → may RETURN here
          |
          ├── self-contained action? → return executor result verbatim
          |
          ├── grounded control action? → grounded synthesis
          |
          └── CHAT → _build_enhanced_system() → broker.infer()/stream
                  → output_governor → response

```

Pipeline stages logged at runtime: Stage 1 Intent → ... → Stage 12 Confidence. Quick mode short-circuits stages 2-4/HyDE; the orchestrator runs all 12.

### 4. Routing layer (eli/execution)

- **router\_enhanced.py** — route(text) -> {action, args, confidence, meta}. Regex-first **priority pipeline** (“explicit priority pipeline installed”): ordered prepasses (web realtime lookup, media, file/PDF, memory, control...) then a core router, with an LLM-intent fallback (cognition/llm\_intent.py). Holds module state like `_last_used_path` for referential follow-ups.

- **execution\_planner.py** / **execution\_intent\_packets.py** — typed ExecutionPlan / RouteDecision artifacts the bus consumes.
- **route\_authority.py**, **route\_contracts.py**, **tool\_execution\_authority.py**, **operator\_policy.py** — guardrails on what may be routed/executed.
- **portable\_intent\_contract.py**, **router\_plugin\_intents.py**, **executor\_plugin\_handlers.py**, **media\_runtime.py**, **operator\_actions.py** — plugin/media/OS intent wiring.

Key router behaviours (each is a fixed bug → guarded by tools/eval/cases.yaml): media controls require **whole-word + command-shape** (no substring “tab” in “metabolise”); bare “do a search” needs a subject; meta-questions stay CHAT; realtime facts (“release date”) → WEB\_SEARCH.

## 5-6. Agent Bus & the agents (eli/cognition/agent\_bus.py)

AgentBus.dispatch(user\_input, intent, ...) -> DispatchResult. Selects an agent set (\_select\_agents\_for\_intent, or broad fan-out), runs them over a dependency DAG (falls back to flat parallel), aggregates.

DispatchResult fields: agent\_results[], action\_result, memory\_context, aggregated\_confidence (route+grounding blend), **grounding\_confidence** (agent-evidence only — the honest “is this backed by anything” signal), agents\_used, confidence\_label, orchestrator\_plan.

**14 registered bus agents** (\_ALL\_AGENTS):

| Agent                 | Role                                                   |
|-----------------------|--------------------------------------------------------|
| BusMemoryAgent        | recall from SQLite/FTS/FAISS into context              |
| KnowledgeGraphAgent   | entities/relations from the KG                         |
| SystemAgent           | runs executor actions (WEB_SEARCH, RUNTIME_AUDIT, ...) |
| OrchestratorAgent     | plan/coordinate grounded synthesis                     |
| FileCodeAgent         | searches the codebase for code-grounded answers        |
| IntrospectionBusAgent | live runtime/self introspection                        |
| CapabilityAgent       | what ELI can do (capability manifest)                  |
| ReflectionAgent       | session reflection/insights                            |
| ProactiveAgent        | proactive patterns/signals                             |
| HabitAgent            | learned habits                                         |
| SelfImprovementAgent  | self-improvement state                                 |
| FrontierAgent         | frontier reasoning hooks                               |
| PluginAgent           | dispatches to registered plugins                       |
| VoiceAgent            | STT/TTS engine state                                   |

+ **the 15th: CodeAgent** (eli/coding/agent.py) — a *separate* coding pipeline (plan → search → verify → repair, beam/DAG), invoked for CODE\_SOLVE / GENERATE\_SCRIPT, not a bus agent.

## 7. The Orchestrator — full 12-stage (eli/cognition/orchestrator.py)

Runs for non-quick modes (and forced cases). Stages (as logged):

1. **Intent Routing** → action
2. **Persona Lock** → persona/identity fixed for the turn
3. **HyDE** → hypothetical-doc query expansion (cognition/hyde.py)
4. **Planner** → which retrieval channels + limits for the mode 5/6/7. **Parallel Retrieval** → keyword + **FTS5** (conversation\_turns) + **FAISS** vector + RAG + **KG**
5. **Hybrid Merge** → combine channel hits
6. **Rerank** → cognition/reranker.py

7. **Context Assembly** → assembled grounded context 10.5. **Persona Handoff** → persona brief built for generation
8. **LLM Generation** → stream/oneshot via the broker
9. **Confidence** → score vs threshold (PASS/repair)

Quick mode (default) skips 2-4/HyDE and uses the AgentBus directly; the orchestrator is the “deep” path.

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## 8. Inference layer (eli/cognition)

- **inference\_broker.py** — the **only canonical inference path**. `broker.infer()`.
- **gguf\_inference.py** — llama-cpp wrapper, live runtime params, `chat_completion()`, runtime snapshot publishing. Model-agnostic; loads whatever GGUF is configured.
- **reasoning\_modes.py** — modes: quick, chain\_of\_thought, self\_consistency, tree\_of\_thoughts, constitutional\_ai (private execution strategy, not shown raw).
- **chat\_model.py**, **context\_builder.py**, **context\_synthesiser.py**, **user\_info\_builder.py** — prompt/context assembly.
- **llm\_intent.py** — LLM fallback for ambiguous routing. Decoding is constrained by a GBNF grammar built from the live action catalogue, so the model can only name a capability that actually exists and can only emit well-formed JSON — an invented action is unrepresentable rather than rejected after the fact. Backends without grammar support (Ollama, remote) fall back to the free-text path unchanged.
- Token/context budgeting lives in `engine.py::_build_enhanced_system` (budget-aware persona trim) + `core/dynamic_runtime_budget.py`.

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## 9. Grounding & anti-confabulation spine (eli/runtime + eli/core/grounding.py)

ELI’s signature subsystem — keeps a local model from confidently stating unbacked facts.

- **deterministic\_grounding\_gate.py** (4.3k) — the gate; decides when an answer must be backed by evidence vs may be free CHAT; blocks raw evidence/telemetry packets from leaking to the user.
- **grounding\_escalation.py** — *tiered* escalation (built this cycle): a checkable factual turn with **low grounding** escalates — external fact → web agent, self/project fact → broad local agent fan-out — and **hedged honestly** if nothing grounds it. Trigger is grounding, *not* the (often-high) response-confidence. Env: `ELI_GROUNDING_ESCALATION`.
- **evidence\_ledger.py**, **evidence\_store.py**, **evidence\_arbitration.py**, **memory\_evidence.py** — what counts as evidence, where it’s recorded, conflict resolution.
- **persistence\_gate.py** — stops internal report-dumps/junk being written to memory (and replayed later).
- **response\_contracts.py**, **response\_packets.py**, **response\_policy.py**, **final\_response\_assembly.py**, **final\_response\_provider.py**, **user\_visible\_response\_surface.py** — shape/clean the final user-visible answer; format internal “surface packets” into readable text (never raw JSON).
- **truth\_report.py**, **eli\_identity\_audit.py**, **control\_contracts.py** — runtime truth evidence, identity grounding, control-action contracts.
- **output\_governor.py**, **response\_governance.py**, **response\_sanitizer.py**, **persona\_hygiene.py** (cognition) — final governance/sanitisation pass.

This is also where the **known weak seam** lives: internal-state→spoken-output leaks (identity confabulation, telemetry dumps) — partially defended, the active frontier (see §20).

## 10. Execution layer (eli/execution/executor\_enhanced.py)

- `execute(action, args)` -> dict — **216 dispatch actions**; runtime capability manifest reports **216 capabilities, 199 of them routable** (*live, read from the manifest each boot*).
- **PHASE45 fast-path** (`engine.py`) — deterministic OS/media/status/job actions (`VOLUME`, `MEDIA_CONTROL`, `NEXT_MEDIA`, `OPEN_APP`, `DATE`, `SHELL_EXEC`, `ANALYZE_IMAGE`, `CHECK_JOB`, `BACKGROUND_JOBS`, ...) bypass the LLM and return the executor result **verbatim** — never paraphrased.
- Action families: media/OS control, file/document (`SUMMARIZE_FILE`, `ANALYZE_PDF[_FOLDER]` incl. per-file mode, `CREATE_DOCUMENT`), web (`WEB_SEARCH`), news, weather, memory (`MEMORY_STORE/RECALL`), code (`CODE_SOLVE`, `GENERATE_SCRIPT` → §coding), background jobs, self/runtime reports (`RUNTIME_STATUS`, `SELF_REPORT`, `RUNTIME_AUDIT`, `EXPLAIN_*`), vision, image generation, scheduling/reminders.
- **Background jobs** (`runtime/background_tasks.py`) — submit/get/list; heavy PDF-folder analysis auto-backgrounds; `CHECK_JOB` surfaces the real result.

## Plugins (eli/plugins, manager.py)

Bundled: `calendar`, `document_reader`, `media`, `notes`, `pomodoro`, `system_stats`, `tts`, `weather`, `web`, `web_automation`. State in `config/plugins_state.json` (gitignored). Custom agents/plugins are trust-gated.

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## 11. Memory subsystem (eli/memory)

- **memory.py** (`Memory`, `get_memory()`) — long-term store over SQLite + **FTS5**, plus failure/observation/habit logging with anti-junk filters.
- **vector\_store.py** — **FAISS** index at `artifacts/vectors/index.faiss`; embedder models/`embeddings/nomic-embed-text-v1.5.Q4_K_M.gguf` (*runtime*).
- **Knowledge graph** — `kg_entities` / `kg_relations` (FTS5-backed) in the user DB.
- **Working memory** (`cognition/working_memory.py`) — session-pinned facts, junk-pin guard; restored across sessions.
- **Persona/personal memory** — `runtime/personal_memory_*`, `persona_updater`.

**Two databases** (*runtime*): `artifacts/db/user.sqlite3` (user-facing) and `artifacts/db/agent.sqlite3` (agent/self-improvement). Observed tables include: `memories(+fts)`, `conversation_turns`, `conversations`, `kg_entities(+fts)`, `kg_relations`, `recall_log`, `runtime_events`, `news_articles(+fts)`, `news_reflections`, `habits/habit_events/habit_rules`, `observations`, `learning_replay`, `working_memory_pins`, `user_patterns`, `session_summaries`, `corrections`, `failures`, `error_tracking`, `improvements`, `capability_proposals`, `code_patches`, `agent_dispatches`, `agent_metrics`. Retrieval is hybrid: keyword + FTS5 + FAISS + RAG + KG, merged & reranked (§7).

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## 12. Perception (eli/perception)

- **Vision** — `vision.py` (model-agnostic VL: Moondream / Qwen2.5-VL hot-swap; mtmd encoder forced to CPU to avoid CUDA clip segfault), `analyze_image.py`, `ambient_vision.py` (ambient toggle), `screen_locator.py`, `gaze_engine.py`, `analyze_mesh.py`, `extract_equations.py`.
  - **STT** — `local_whisper_stt.py` (faster-whisper `small.en`, CPU int8, local\_only when offline), `audio_stt.py`, `eli_listen.py`, `voice_worker*.py` (wake-word “computer”, music-bleed filter, echo gate).
  - **TTS** — `tts_router.py` (Piper voices, e.g. `en_US-amy-medium`).
  - **OS** — `os_controller.py` (app/window/keyboard/mouse), `analyze_csv.py`, `analyze_pdfs.py`.
-

### 13. Persona system (eli/cognition/persona\*)

- `persona.txt` (authored) + `persona.auto.txt` (overlay) — the voice.
  - `persona_updater.py` — updates overlay/KG/user-profile each turn (tone signals).
  - `persona.py`, `persona_values.py`, `persona_status.py`, `persona_hygiene.py`.
  - `tone_analyzer.py` — writes tone preference signals (correction/humor/depth).
  - Generation injects a budget-trimmed persona + situation brief (`engine.py::_build_enhanced_system`); a live-runtime fact is injected for model/identity questions so ELI reports the loaded model instead of denying it.
- 

### 14. Daemons & background loops

- **Proactive daemon** (`planning/proactive_daemon.py`) — continuous learning; pattern signals (time habit, topic focus, recurring errors, active project).
  - **Self-improvement loop** (`runtime/self_improvement.py`) — learns from logged failures (now filtered of transient/user errors).
  - **Habit scheduler** (`planning/habits_scheduler.py`, `habits.py`) — learned routines; guarded against junk-rule replay.
  - **Scheduler / task bus** (`kernel/scheduler.py`, `kernel/task_bus.py`).
  - **Background tasks** (`runtime/background_tasks.py`) — async heavy jobs.
  - **Code monitor** (`runtime/code_monitor.py`), **ambient vision loop**.
  - **World event bus** (`world/world_event_bus.py`) — fed confidence/agent events.
- 

### 15. Learning / self-training (eli/learning)

LoRA fine-tuning pipeline on a Phi-3 base: `bootstrap_phi3_base.py`, `base_model_resolver.py`, `dataset_builder.py`, `dataset_filters.py`, `merge_reviewed_datasets.py`, `export_trainable_dataset.py`, `training_preflight.py`, `lora_trainer.py`, `lora_trainer_guard.py`, `lora_eval.py`. Turns reviewed interactions into trainable datasets and adapters.

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### 16. EliWorld (eli/world, eli/gui EliWorld tab, kernel/world\_model.py)

An internal/spatial “world” the agent inhabits (rooms like the Reflection Chamber/Anomaly Room appear (*runtime*)). `world_event_bus.py` receives runtime events; `local_world_bridge.py` bridges to the model; snapshots/ledger/journal under `artifacts/world/`. Experimental/creative subsystem.

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### 17. Core infrastructure (eli/core)

- **netguard.py** — offline-by-default: `guarded_urlopen`, process-wide socket guard (fail-closed on non-loopback while offline), `allow_network()` scoped context for deliberate user-initiated fetches (e.g. model download).
- **paths.py**, **portable\_paths.py**, **legacy\_paths.py**, **db\_paths.py** — canonical path resolution (dev vs platformdirs; `ELI_*` overrides).
- **runtime\_settings.py**, **config.py** — settings load/merge/heal; DEFAULTS are clean (offline, no model, wizard on). `config/settings.json` is per-user/gitignored, seeded from `config/templates/settings.template.json`.
- **hardware\_profile.py**, **startup\_hardware\_optimizer.py**, **dynamic\_runtime\_budget.py** — detect GPU/VRAM, pick ctx/gpu\_layers/batch.
- **first\_run.py**, **first\_run\_wizard.py** — onboarding state.



- **model\_download.py** — curated GGUF downloader (catalog, resumable, GGUF-magic + size validated, netguard-gated). Install-time menu only — inference stays model-agnostic.
- **grounding.py** — `is_grounded_query` etc. (shared grounding helpers).

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## 18. GUI (eli/gui)

PySide6 (LGPL — not PyQt/GPL, see `pyproject.toml`). `eli_pro_audio_gui_v2_0.py` (main window, 11k LOC), `app.py` (entry/boot), `labs_tab.py` (Experimental), panels in `eli/gui/panels/` (`startup.py` = `StartupModelSelectionDialog` + `FirstBootWizard` with curated download, `HardwareTuningDock`). `EliWorld` tab, Operator console, Proactive dock. `qt_compat.py` shim allows PyQt fallback from source.

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## 19. On-disk layout (artifacts/, runtime)

```
artifacts/
├─ db/{user,agent}.sqlite3      # memory + agent state
├─ vectors/index.faiss         # semantic index
├─ conversations/ , conversations/archive/
├─ incidents/<date>.jsonl      # incident log
├─ proactive/ , world/{snapshots,ledger,journal}/
├─ runtime/users/<uuid>/user_profile... # per-user profile
├─ runtime_snapshot.json       # live model/runtime truth
├─ documents/ , scripts/      # generated artifacts
├─ analyze_image_*/           # vision outputs
config/  settings.json(gitignored) · templates/ · plugins_state.json ...
models/  <your>.gguf · embeddings/ · whisper/ · image/
voices/  Piper ONNX
```

---

## 20. Known seams & weak points (honest)

1. **God-files** (§1) — four files  $\approx$  a third of the codebase; the active refactor target. Split by concern, hold a size ceiling, keep a repair-log.
2. **Internal-state  $\rightarrow$  spoken-output seam** — the recurring failure class (identity confabulation, telemetry/packet leaks, ungrounded factual claims). The grounding gate + escalation defend it but don't fully close it on the plain CHAT path. This is *the* frontier item.
3. **Swallowed errors** — many except `Exception: pass`; convert to `logged/raised` so failures surface instead of hiding.
4. **No capability eval until now** — `tools/eval/` (see `tools/eval/README.md`) is the new measurement layer; coverage grows by turning every bug-log into a case.
5. **Latency vs model size** — on an 8GB GPU a 24B at Q5 offloads few layers  $\rightarrow$  minutes/reply; a 7-9B Q4 is the sweet spot. The model picker should warn on poor VRAM fit.

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## 21. Where to make a change (quick index)

| To change...                     | Edit...                                         |
|----------------------------------|-------------------------------------------------|
| how input is routed to an action | <code>eli/execution/router_enhanced.py</code>   |
| what an action does              | <code>eli/execution/executor_enhanced.py</code> |
| which agents run / how grounded  | <code>eli/cognition/agent_bus.py</code>         |
| the deep 12-stage retrieval      | <code>eli/cognition/orchestrator.py</code>      |

| To change...                                                                    | Edit...                                                                                         |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| prompt/persona budget, the engine spine<br>anti-confabulation / verify-or-hedge | eli/kernel/engine.py<br>eli/runtime/grounding_escalation.py,<br>deterministic_grounding_gate.py |
| final answer shaping/cleanup                                                    | eli/runtime/*response*,<br>cognition/output_governor.py                                         |
| memory store/recall                                                             | eli/memory/memory.py, vector_store.py                                                           |
| model loading / inference                                                       | eli/cognition/gguf_inference.py,<br>inference_broker.py                                         |
| offline/network policy                                                          | eli/core/netguard.py                                                                            |
| paths / settings / model download                                               | eli/core/{paths, runtime_settings, model_download}.py                                           |
| the GUI                                                                         | eli/gui/eli_pro_audio_gui_v2_0.py, gui/panels/                                                  |
| eval / regression board                                                         | tools/eval/ (+ tools/eval/README.md)                                                            |

““